

Supporting Information for Random phase approximation-based local natural orbital coupled cluster theory

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I. BASIS SETS AND GEOMETRIES

The solid-optimized cc-pVTZ basis sets for Cu, as well as the fitting basis sets for Cu and Li used in this work, are available in the following GitHub repository:

github.com/hongzhouye/supporting_data/tree/main/2026/RPALNOCC

The same repository also contains the structure files for the C2C2PD molecular dimer and the anthracene molecular crystal, including all geometries required for counterpoise corrections.

II. SUPPLEMENTARY FIGURES

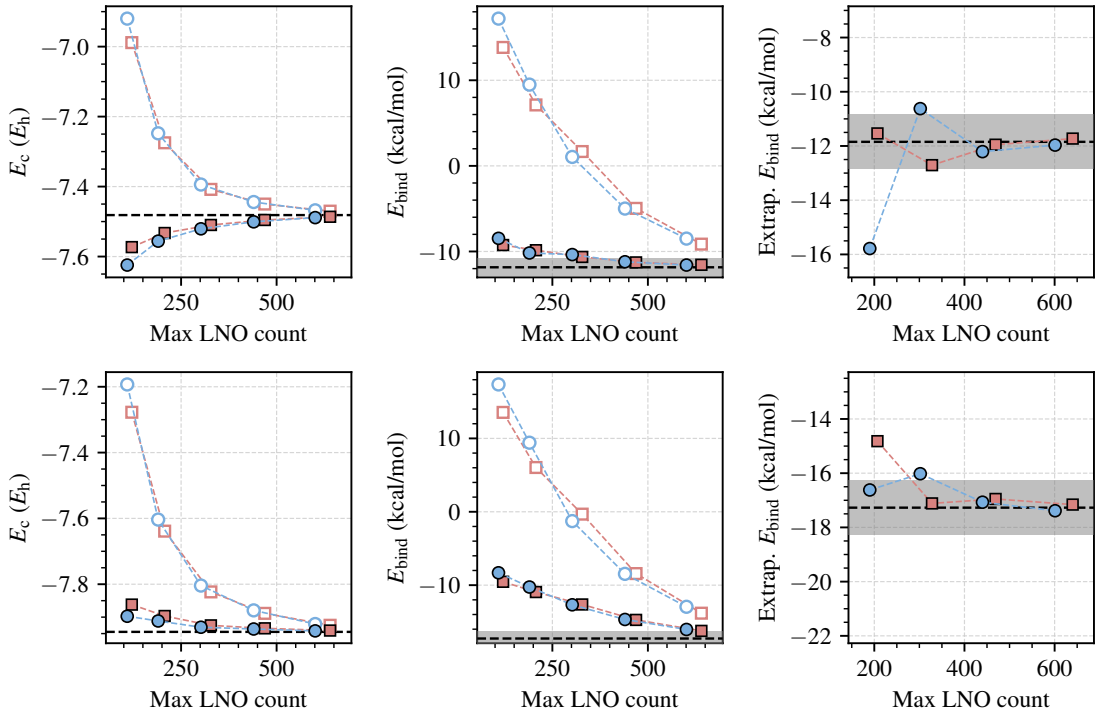


FIG. S1. LNO convergence results corresponding to FIG. 2 in the main text, but comparing RPA-based dLNOs (red) with RPA-based LNOs (blue). The results for the RPA-based LNOs are identical to those shown in FIG. 2 of the main text.

Alt text: Plots of LNO-CC correlation and binding energies for the C2C2PD complex as a function of LAS size (maximum number of active LNOs), comparing RPA-based LNO-CC using normal and direct LNO definitions.

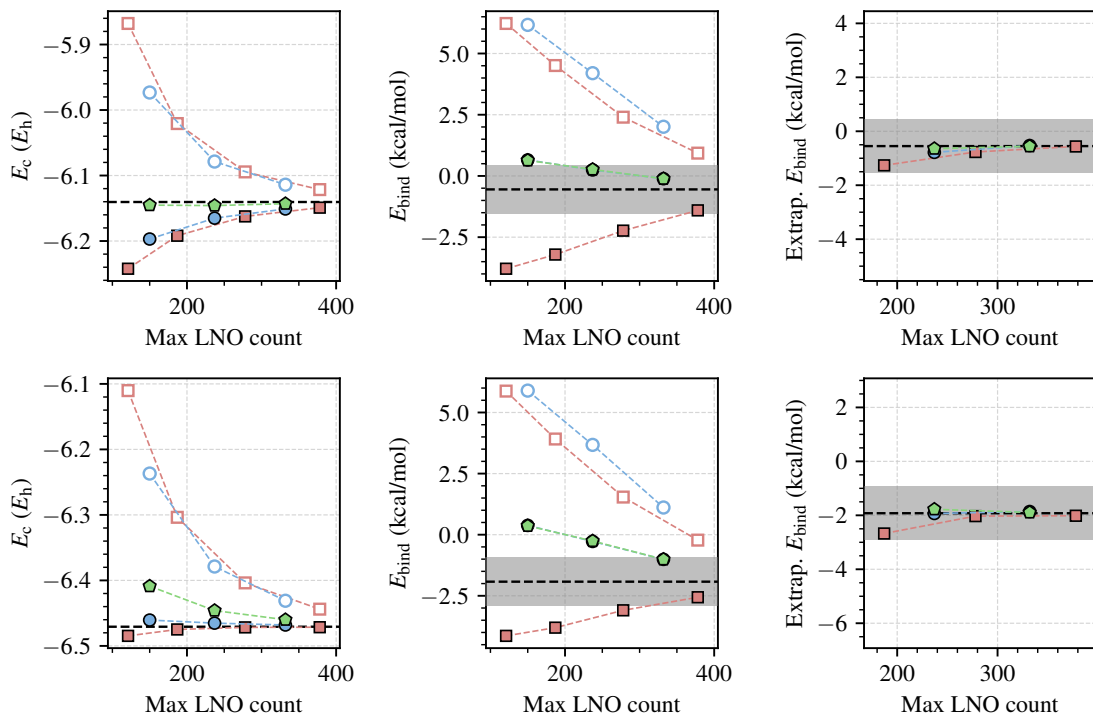


FIG. S2. LNO convergence results corresponding to FIG. 2 in the main text, but for GGG in the aug-cc-pVTZ basis sets. Red, MP2-LNO-CC; blue, RPA-LNO-CC; green, SOSEX-LNO-CC.

Alt text: Plots of LNO-CC correlation and binding energies for the GGG molecular complex as a function of LAS size (maximum number of active LNOs), comparing MP2-, RPA-, and SOSEX-based approaches.

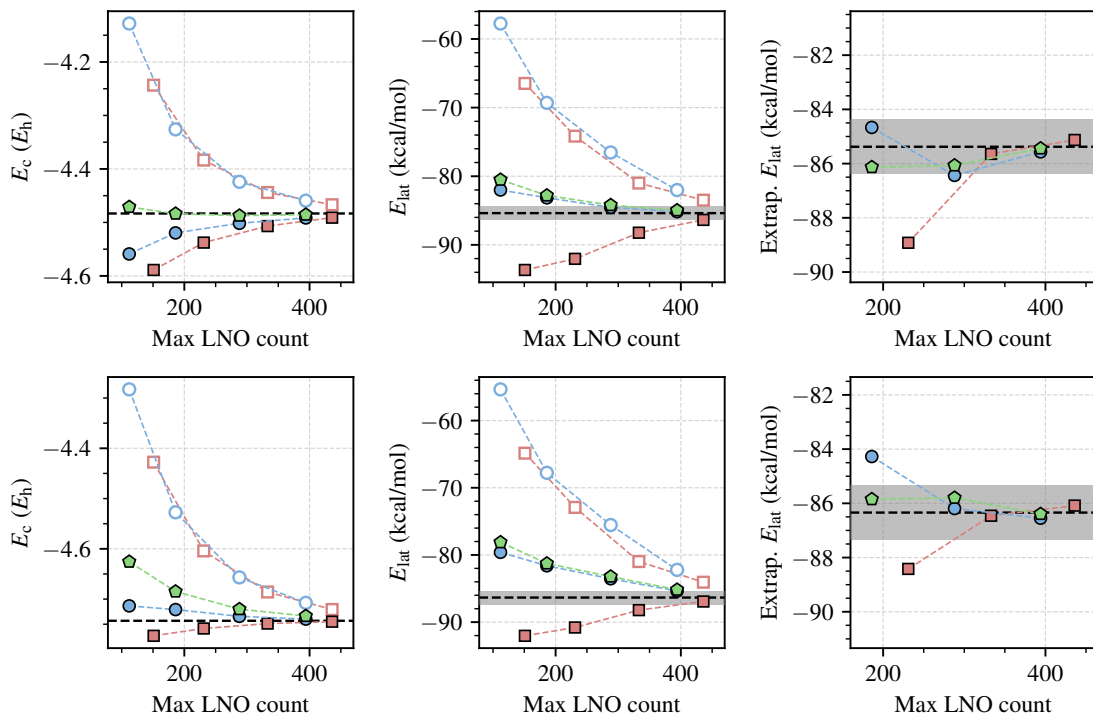


FIG. S3. LNO convergence results corresponding to FIG. 2 in the main text, but for the anthracene crystal in the cc-pVTZ basis sets with Γ -point Brillouin zone sampling. Red, MP2-LNO-CC; blue, RPA-LNO-CC; green, SOSEX-LNO-CC.

Alt text: Plots of LNO-CC correlation and binding energies for the anthracene molecular crystal as a function of LAS size (maximum number of active LNOs), comparing MP2-, RPA-, and SOSEX-based approaches.

III. SUPPLEMENTARY DATA

TABLE S1. Computational cost of the individual steps in an RPA-based LNO-CCSD(T) calculation for C2C2PD using the cc-pVTZ basis sets and an LNO truncation threshold of $\eta_{\text{vir}} = 3 \times 10^{-7}$. The drCCD equations are solved only once, whereas the reported wall times for the RPA-based LNO construction and the fragment CC calculations are accumulated over all 108 fragments of this molecule. In practice, the fragment calculations are carried out in parallel, with each fragment assigned 24 EPYC 7763 CPU cores and 96 GB of total memory.

Alt text: Table of LNO-CC cost breakdown by individual steps for the C2C2PD molecular complex, including drCCD equation, RPA LNO construction, fragment CCSD, and fragment (T).

Step	Wall time (hour)
drCCD equation	1.1
RPA LNO construction	5.5
Fragment CCSD	943
Fragment (T)	3750

TABLE S2. Per-atom correlation energy (E_h) calculated by different methods for BCC Li and FCC Cu using different basis sets and k -point meshes. The CCSD results for $2 \times 2 \times 2/\text{DZ}$ and $3 \times 3 \times 3/\text{DZ}$ are from canonical k -point CCSD, while all others are the best extrapolated values from LNO-CCSD as explained in the main text.

Alt text: Table of LNO-CCSD correlation energies (E_h) for BCC lithium and FCC copper across different k -point meshes and basis sets, comparing MP2-, RPA-, and SOSEX-based LNO-CCSD results with canonical CCSD.

System	N_k	Basis set	MP2	RPA	RPA+SOSEX	CCSD
BCC Li	$3 \times 3 \times 3$	TZ	-0.0604	-0.0862	-0.0488	-0.0630
	$2 \times 2 \times 2$	DZ	-0.0344	-0.0478	-0.0276	-0.0388
BCC Li	$3 \times 3 \times 3$	DZ	-0.0463	-0.0535	-0.0325	-0.0434
	$4 \times 4 \times 4$	DZ	-0.0534	-0.0557	-0.0346	-0.0452
FCC Cu	$2 \times 2 \times 2$	TZ	-0.5583	-0.4823	-0.3789	-0.4690